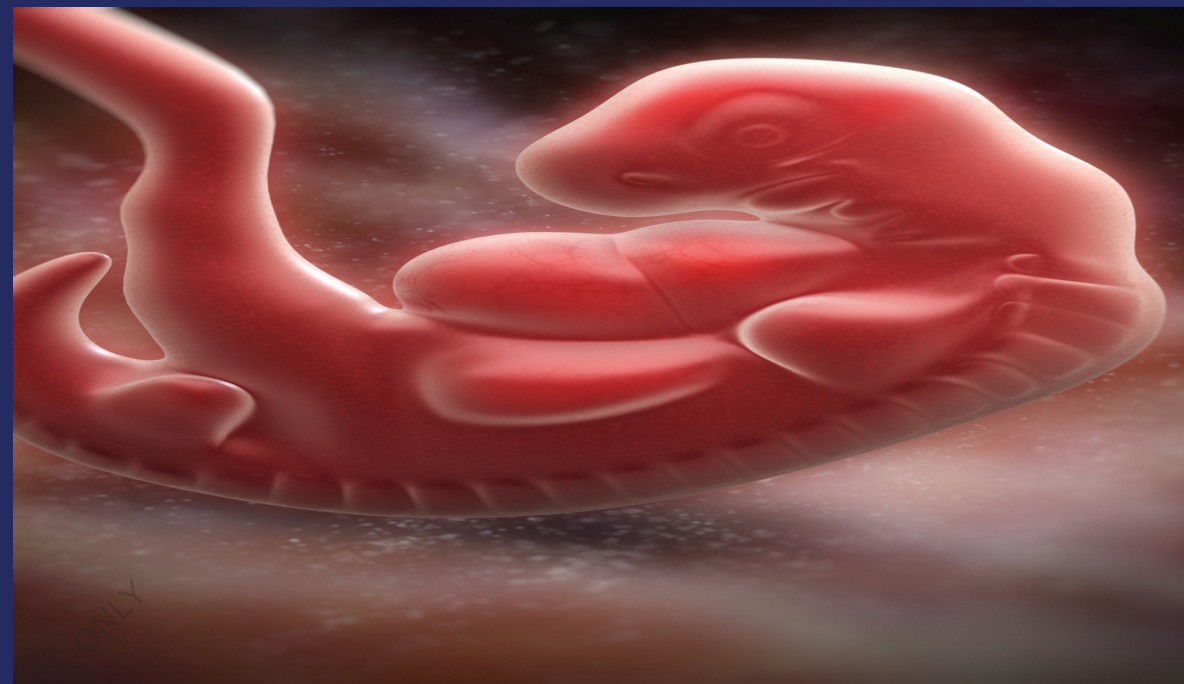


The practical manual adheres to the scope and sequence of most major courses nationwide and strives to make laboratory biology, as a discipline, interesting and accessible to students. In addition to comprehensive coverage of core concepts and foundational research, it incorporates features that engage learners with the discipline in meaningful ways. This practical manual provides fundamental concepts of laboratory biology with a scope that is manageable for both instructors and students.



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The Manual of Cytology, Genetics and Developmental Biology

Lab Course-II. A Revised Syllabus based on Periyar
University



**Kalaimani Subramani
Kandeepan Chithan
Umamaheswari Subramani**

The Manual of Cytology, Genetics and Developmental Biology

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**THE MANUAL OF
CYTOLOGY, GENETICS AND DEVELOPMENTAL BIOLOGY
(LAB COURSE-II)**

(A Revised Syllabus based on Periyar University)

2025

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FOREWORD

The Practical Manual adheres to the scope and sequence of most one major course nationwide. also strive to make Lab biology, as a discipline, interesting and accessible to students. in addition to a comprehensive coverage of core concepts and foundational research, incorporated features that draw learners into the discipline in meaningful ways. this practical manual provides a fundamental concept of Lab Biology with a scope that is manageable for instructors and students alike with this; it is my pleasure to write a brief foreword to this wonderful Book on “**THE MANUAL OF CYTOLOGY, GENETICS AND DEVELOPMENTAL BIOLOGY (LAB COURSE-II)**” I am appreciating the authors to chosen this manual chapter on basic preparatory knowledge of Lab course in biology. Number of Book are available in the form of hand book, question bank for competitive examinations etc. except from this for the student community limited numbers of this like preparatory book are available.

UG or PG level lack of syllabus covered Book are available, this Manual might be useful to basic and advanced learner of Lab biology. written of this practical book contain simple language and easy to understand a basic learner. this Book offers a valuable guide to the intellectual. this practical manual can help teachers develop a shared vision and understanding of basic concepts of biology and techniques of biology and its flexible uses across subjects and grade levels.

This Practical Mnual has been created with several goals in mind, accessibility, customization, and student engagement all while encouraging students toward high levels of academic scholarship. instructors and students alike will find that this preparatory manual offers a strong introduction to Lab biology in an accessible format.

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Dr. S. KALAIMANI

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MAJOR PRACTICAL

1.COUNTING RBC & WBC BLOOD CELLS USING A HAEMOCYTOMETER

AIM

Counting of RBC and WBC using Haemocytometer

MATERIALS NEEDED

- Haemocytometer (usually Neubauer chamber)
- Red blood cell (RBC) pipette & diluting fluid (e.g., Hayem's or Gower's solution)
- White blood cell (WBC) pipette & diluting fluid (usually acetic acid or Turk's solution)
- Blood sample (capillary or EDTA-anticoagulated)
- Coverslip (special for haemocytometer)
- Microscope
- Cotton with alcohol (for cleaning)

A. Counting RBCs (Red Blood Cells)

PROCEDURE:

1. Clean Equipment: Ensure the haemocytometer and coverslip are clean. Wipe with alcohol and dry.
2. Blood Collection: Take blood up to 0.5 mark in an RBC pipette, avoiding air bubbles.
3. Dilution: Fill the pipette with RBC diluting fluid (up to 101 mark) to achieve a 1:200 dilution. Mix thoroughly by rotating horizontally.
4. Load Chamber:
 - Discard the first 2–3 drops.
 - Place the coverslip on the haemocytometer platform.
 - Gently charge the chamber by touching the pipette tip at the edge of the coverslip, allowing fluid to fill by capillary action without air bubbles.
5. Let Settle: Allow cells to settle for 2–3 minutes.
6. Microscopy:
 - Focus on the grid using low (then high) power.
 - For RBC count: Use the central grid, count all cells in five groups of 16 small squares (four corners + center of the central square, total 80 small squares).
7. Counting Rule: Count cells touching top and left borders; exclude those on right and bottom borders (to avoid counting twice).

8. Calculation: Use the formula:

FORMULA

$$\text{RBC count} = \frac{\text{No. of cells counted} \times \text{dilution factor}}{\text{volume counted}}$$

For a standard Neubauer chamber, volume for 80 squares is 0.02 mm^3 .

B. Counting WBCs (White Blood Cells)

PROCEDURE:

1. Prepare Haemocytometer: Clean chamber and coverslip thoroughly.
2. Blood Collection: Draw blood up to the 0.5 mark of a WBC pipette, avoiding air bubbles.
3. Dilution: Add WBC diluting fluid up to the 11 marks for a 1:20 dilution. Mix by rotating the pipette horizontally.
4. Load Chamber:
 - Discard the first few drops.
 - Fill the chamber as with RBCs, ensuring no bubbles.
5. Settle: Let the WBCs settle for a few minutes.
6. Microscopy:
 - Use low power (10x).
 - For WBC count: Count cells in the four large corner squares of the haemocytometer grid—each large square contains 16 smaller squares, but WBCs are counted in the large ones only.
7. Counting Rule: Apply the same rule as for RBCs—count cells on top and left borders, not those touching right and bottom.
8. Calculation: Use the formula:

$$\text{WBC count} = \frac{\text{No. of cells counted} \times \text{Dilution factor}}{\text{Volume counted}}$$

Clean Up

- Clean the haemocytometer and coverslip thoroughly before storage or reuse.

Volume counted

With a dilution of 1:20, and each large square having a known volume

COUNTING OF WBC

LUC - I

3	4	5	7
6	8	4	6
5	4	9	2
9	9	7	6

RUC - II

6	7	8	9
5	7	6	6
8	5	3	11
8	10	6	1

LIC - III

8	12	10	1
2	7	8	5
9	10	11	7
8	6	5	11

RLC - IV

12	10	5	2
3	0	5	6
9	10	8	5
6	7	8	9

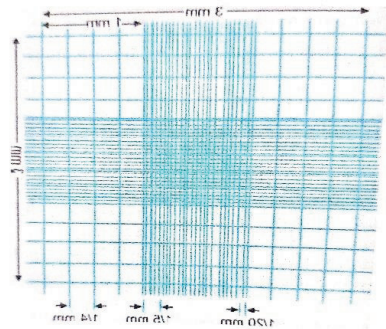
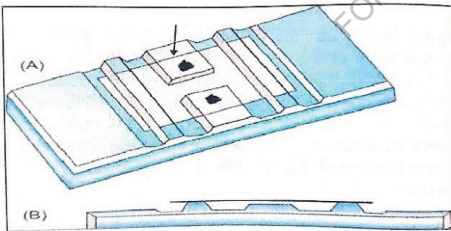
Total number of cells counted in 5 large cells of WBC

LUC - I	RUC - II	LIC - III	RLC - IV	Total
89	106	120	109	424

Calculation:

$$\begin{aligned}
 \text{Number of cell (N)} &= 424 \\
 \text{Area counted} &= 1 \text{ mm}^2 \times 4 = 4 \text{ mm}^2 \\
 \text{Depth} &= 1/10 \text{ mm} \\
 \text{Dilution} &= 1:20 = 1/20 \\
 \text{WBC / cu.m} &= N \times 50 / \text{Cu. mm} \\
 &= 424 \times 50 \\
 &= 21,200 / \text{Cu. mm}
 \end{aligned}$$

$$\frac{424}{4} =$$



2.PREPARATION AND IDENTIFICATION OF SLIDES OF MITOTIC DIVISIONS WITH ONION ROOT TIPS.

Aim

To study and demonstrate mitosis by preparing the mount of an onion root tip cell.

Materials Required

- Compound microscope
- Acetocarmine stain
- Water
- Burner
- N/10 Hydrochloric acid
- Filter paper
- Coverslip
- Aceto alcohol (Glacial acetic acid and Ethanol in the ratio 1:3)
- Glass Slide
- Onion root peel
- Forceps
- Blade
- Watch glass
- Dropper
- Needle
- Vial

Theory Of the Experiment

It is because of the meristematic cells that are situated in the tip of the roots that render the most desirable and suitable raw material to study the different stages of mitosis. Onion is a monocot plant. Monocotyledonous plants possess large chromosomes that are clearly visible. Hence, their root tips are used. The period of time taken for mitosis varies as it is dependent on the cell type and type of species.

Procedure Of the Experiment

- Place an onion on a tile
- With the help of a sharp blade, carefully snip the dry roots of the onion
- Place the bulbs in a beaker containing water to grow the root tips
- It may take around 4 to 6 days for the new roots to grow and appear
- Trim around 3 cm of the newly grown roots and place them in a watch glass
- With the help of forceps, shift it to a vial holding freshly prepared aceto-alcohol i.e., a mixture of glacial acetic acid and ethanol in the ratio 1:3
- Allow the root tips to remain in the vial for one complete day
- With the help of forceps, pick one root and set in on a new glass slide
- With the help of a dropper, allow one drop of N/10 HCl to come in contact with the tip of the root. Additionally, add around 2 to 3 drops of the acetocarmine stain
- Heat it lightly on the burner in such a way that the stain does not dry up
- Excessive stain can be carefully treated using filter paper
- The more stained part of the root tip can be trimmed with the help of a blade.
- Discard the lesser stained part while retaining the more stained section
- Add a droplet of water to it
- With the help of a needle, a coverslip can be mounted on it

- Gently tap the coverslip with an unsharpened end of a needle in order for the meristematic tissue of the root tip present under the coverslip to be squashed properly and to be straightened out as a fine cell layer
- The onion root tip cells' slide is now prepared and ready to be examined for different stages of mitosis
- Observe and study mitosis by placing the slide under the compound microscope. Focus as desired to obtain a distinct and clear image.

Observations and Conclusion

- The slide containing the stained root tip cells is placed on the stage of the compound microscope, changes taking place are noted and sketched.
- The different phases of mitosis, such as prophase, metaphase, anaphase and telophase can be observed.

Stages Of Mitosis

1. Prophase

- The process of mitosis is initiated at this stage wherein coiling and thickening of the chromosomes occurs
- Shrinking and hence the disappearance of the nucleolus and nuclear membrane takes place
- The stage reaches its final state when a cluster of fibres organizes to form the spindle fibres

2. Metaphase

- Chromosomes turn thick in this phase. The two chromatids from each of the chromosomes appear distinct
- Each of the chromosomes is fastened to the spindle fibres located on its controller
- Chromosomes align at the centreline of the cell

3. Anaphase

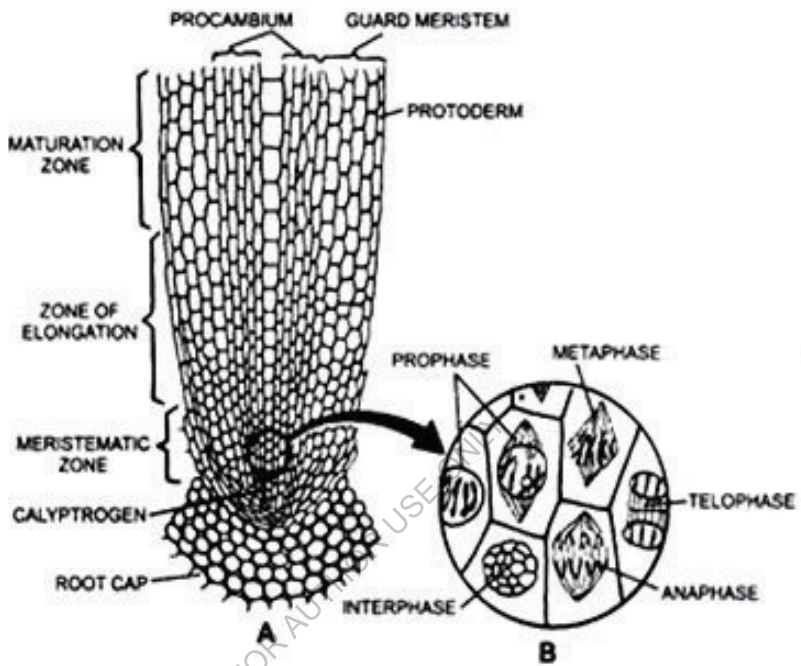
- Each of the chromatid pair detaches from the centromere and approaches the other end of the cell through the spindle fibre
- At this stage, compressing of the cell membrane at the centre takes place

4. Telophase

- Chromatids have reached the other end of the cell
- The disappearance of the spindles
- Chromatin fibres are formed as a result of uncoiling of daughter chromosomes
- The appearance of two daughter nuclei at the opposing ends due to the reformation of the nucleolus and nuclear membrane
- At this phase, splitting of the cell or cytokinesis may also occur

Post mitosis, the next stage is referred to as interphase, which is part of the cell cycle that is non-dividing and between two consecutive cell divisions. A cell spends most of its life in the interphase. It comprises the G1, S and G2 stages.

VEDEO SOURCE : <https://youtu.be/5-ur7bWqlDQ>



3.STAINING AND OBSERVATION OF POLYTENE

CHROMOSOMES IN SALIVARY GLANDS OF CHIRONOMUS

LARVA

AIM

To Staining and observing polytene chromosomes in the salivary glands of Chironomus larvae.

MATERIALS REQUIRED

- Acetocarmine and acetoorcein,
- Slide and Cover glass,
- Immersion oil,
- Magnifying Lense or Dissection microscope,
- Forceps and Dropper,
- Blotting papers

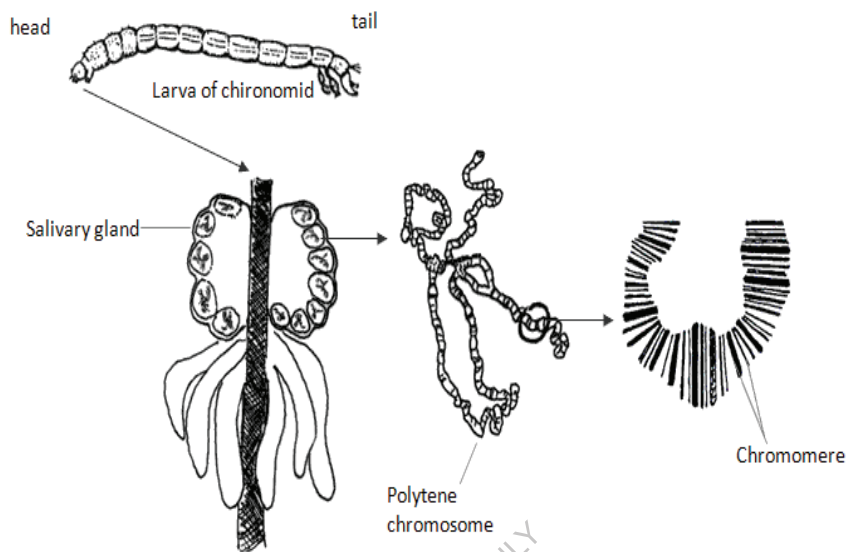
STAINING PROCEDURE

Based on the provided search results, here are the key steps for staining and observing polytene chromosomes in the salivary glands of Chironomus larvae:

1. Collect fresh Chironomus larvae from their natural habitat, such as stagnant pools of water
2. Dissect the larvae under a magnifying lens or dissecting microscope to isolate the salivary glands
3. Place the salivary glands on a glass slide and add a few drops of acetyl carmine or acetocormine stain
4. Gently lower a coverslip onto the slide and tap it a few times with forceps to spread the cells
5. Blot away any excess stain from the edges of the coverslip

MICROSCOPIC OBSERVATION

1. Scan the slide under 10x magnification to locate the salivary gland cells containing the polytene chromosomes
2. Observe the polytene chromosomes under high power (40x, 100x) magnification
3. The polytene chromosomes will appear as alternating dark bands and lightly stained inter bands.
4. The dark bands contain more DNA and less RNA, while the inter bands contain more RNA and less DNA.
5. Enlarged regions called "puffs" may be visible, indicating sites of active gene transcription.
6. The chromatin fibres of the puffs may form lateral loops called "Balbiani rings", which are sites of active RNA synthesis



<https://youtu.be/mh7i3nhzXzk>

4. MOUNTING OF DIFFERENT DEVELOPMENTAL STAGES OF CHICK EMBRYO:24 HR,48HR,72HR AND 96HR

AIM

The practical procedure for mounting different developmental stages of chick embryo (24 hr, 48 hr, 72 hr, and 96 hr) whole mounts involves careful extraction, preparation, and observation of the embryo at each time point.

MATERIALS REQUIRED

- Incubated fertilized eggs (24, 48, 72, 96 hr incubation)
- Dissecting microscope
- Fine forceps and scissors
- Glass rings or nitrocellulose/Whatman paper rings for support
- Saline solution (e.g., phosphate-buffered saline)
- Watch glass or culture dish
- Glass slides and cover slips
- Staining reagents (e.g., methylene blue, if staining is desired)
- Incubator (set to 38°C)

PRACTICAL PROCEDURE

1. Incubation of Eggs

- Incubate fertilized eggs under optimal conditions at 38°C for required durations: 24 hr, 48 hr, 72 hr, and 96 hr to obtain embryos at respective developmental stages.

2. Opening the Egg and Embryo Explanation

- Carefully tap and crack the egg without disturbing the yolk.
- Remove a portion of the eggshell over the air cell side.
- Using fine forceps, gently separate the thin albumin to expose the blastoderm (embryo) on the surface of the yolk.
- Remove excess albumin surrounding the embryo.

3. Isolation of the Embryo

- Tilt the yolk sac to place the embryo facing upward.
- Using fine scissors/forceps, cut the yolk sac around the equator (below the embryo).
- Peel off the vitelline membrane around the embryo gently; ensure it remains intact.
- Place the vitelline membrane with embryo on a watch glass or small culture dish with some saline.

4. Mounting the Embryo on a Glass Ring or Support

- Centre the embryo with the vitelline membrane over a glass ring or a Whatman paper ring.
- Carefully lift the edges of the vitelline membrane to adhere around the ring.
- Trim excess vitelline membrane, avoiding puncture or tearing.
- If culture is needed, the ring with vitelline membrane and embryo can be transferred to a culture dish with thin albumin or saline for observation or incubation.

5. Fixation and Staining (Optional)

- Fix the embryo in appropriate fixative (e.g., 4% paraformaldehyde) for detailed microscopic study.
- Stain whole mounts with dyes like methylene blue or neutral red if visualization of morphological features is required.

6. Observation Under Microscope

- Place the mounted embryo on a glass slide.
- Use a dissecting or compound microscope to observe specific features:
 - *At 24 hr*: Primitive streak, four somite pairs, neural folds, and early notochord formation.
 - *At 48 hr*: Head torsion (rotation), flexion, somite pairs increase, developing brain regions (telencephalon, mesencephalon), and early optic vesicle formation.
 - *At 72 hr*: Optic cup and lens visible, auditory structures develop, extraembryonic membranes distinguishable.
 - *At 96 hr*: Distinct wing and leg buds, 41 pairs of somites, visceral arches, body folds creating C-shaped embryo, left side lying on yolk.

7. Documentation

- Take careful notes or photographic records of each developmental stage for comparative studies.

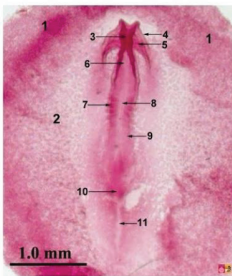
Important Notes:

- Embryos at earlier stages (24–48 hr) are fragile; handling must be very gentle.
- Use humid chambers or maintain moisture to prevent embryo desiccation during mounting.
- For live culture, maintain temperature at 38°C and adequate humidity.

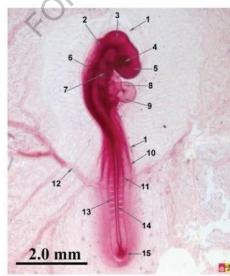
This procedure enables visualization of chick embryogenesis at crucial stages and is essential for studying morphogenesis, organogenesis, and effects of experimental treatments on development.



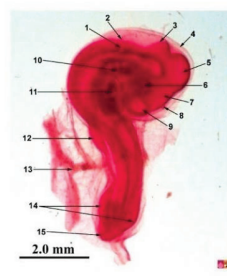
CHICK EMBRYO



24 HOURS



48 HOURS



72 HOURS

MINOR PRACTICAL

FOR AUTHOR USE ONLY

5. MOUNTING OF BUCCAL EPITHELIUM (BARR BODY)

PREPARATION.

AIM

To prepare and mount a slide for observing Barr bodies in buccal epithelial cells

Materials Required:

- Methylene blue stain
- Glass Slide
- Cover glass
- Sealing wax or nail polish

PROCEDURE:

1. Prepare the Slide:
 - Take a clean glass slide.
 - Place a drop of water in the centre of the slide.
2. Scrape the Cheek Cells:
 - Gently scrape the inside of the cheek of a human female using the flat side of a toothpick. This will collect the buccal epithelial cells.
3. Spread the Cells:
 - Spread the oral smear in the water on the slide.
4. Stain the Cells:
 - Add a drop of methylene blue stain (a vital dye) to the slide. This stain will highlight the Barr bodies.
5. Mount the Coverslip:
 - Place a coverslip on the slide and seal the ends with nail paint to prevent the stain from drying out.
6. Observe Under a Microscope:
 - Use a light microscope to observe the squamous epithelial cells.
 - Focus on the nucleus of the cell under high power to observe the Barr body.
7. Repeat for Control:
 - Repeat the same procedure with the oral smear of male students to serve as a control.

OBSERVATION

Barr Body: A darkly stained body just below the nuclear membrane in about half the cells. This is the Barr body, which is not present in all cells but in about half the cells of females.

INFERENCE

The presence of Barr bodies indicates the presence of inactive X chromosomes, which is essential for understanding sex chromatin and dosage compensation in females.

Source

<https://youtu.be/hLt884HV8bE>



XX female	X ← active X • ← Barr body
XY male	X ← active X (the only X) Y
XXY male (Klinefelter)	X ← active X Y • ← Barr body
XXX female (triple X)	X ← active X • • ← Barr bodies

6.BLOOD GROUP DETERMINATION INCLUDING ABO AND Rh

AIM

Blood group determination including ABO and Rh

MATERIALS NEEDED

- Clean glass slide or white porcelain tile
- Monoclonal antibodies: Anti-A, Anti-B, and Anti-D (for Rh factor)
- Lancet or finger-pricking device
- Alcohol swabs
- Sterile cotton balls
- Toothpicks or wooden stirrers
- Biohazard disposal container

STEPWISE PROCEDURE

1. Preparation of the Slide

- Draw three separate circles on the glass slide and label them as Anti-A, Anti-B, and Anti-D.

2. Applying the Reagents

- Place a drop of Anti-A serum in the first circle, a drop of Anti-B serum in the second circle, and a drop of Anti-D serum in the third circle.

3. Sample Collection

- Clean the fingertip (usually ring finger) with an alcohol swab and allow it to dry.
- Prick the finger gently using the lancet.
- Wipe off the first drop of blood with a sterile cotton ball.
- Allow drops of fresh blood to fall onto each of the three marked circles on the slide.

4. Mixing

- Using separate toothpicks for each circle, mix the blood thoroughly with the corresponding antibody.

5. Observation

- Wait for about 1–2 minutes and observe each circle for agglutination (clumping).
 - Agglutination in Anti-A: Presence of A antigen → Blood type has A antigen.
 - Agglutination in Anti-B: Presence of B antigen → Blood type has B antigen.
 - Agglutination in both Anti-A and Anti-B: Blood type AB.
 - No agglutination in Anti-A and Anti-B: Blood type O.
 - Agglutination in Anti-D: Indicates presence of Rh (D) antigen → Rh positive.
 - No agglutination in Anti-D: Rh negative.

6. Cleaning Up

- Safely dispose of lancet, cotton balls, toothpicks, and slide in a biohazard container.

INTERPRETATION CHART:

Agglutination Reaction	Blood Group Determination
Anti-A only	Group A
Anti-B only	Group B
Anti-A and Anti-B	Group AB
Neither Anti-A or Anti-B	Group O
Agglutination in Anti-D	Rh Positive (Rh+)
No agglutination in Anti-D	Rh Negative (Rh-)

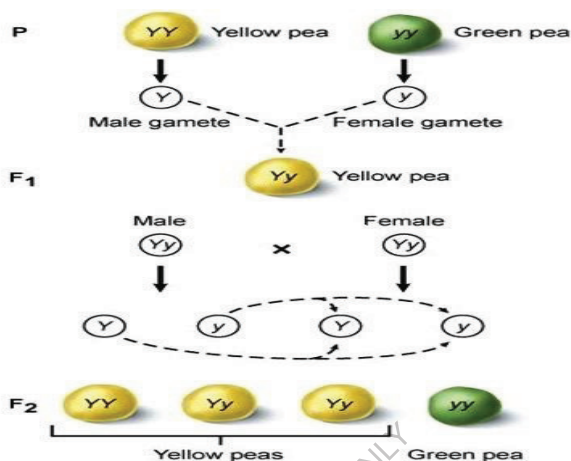
Example: If blood clumps in Anti-A and Anti-D but not in Anti-B, the blood group is A positive (A+).

Key Points

- No special patient preparation is required.
- The test is essential before blood transfusions to avoid immune reactions.
- The presence or absence of Rh factor is critical, especially in pregnancy and transfusion compatibility.
- Blood typing includes forward typing (testing red cell antigens with Anti-A, Anti-B, Anti-D) and sometimes reverse typing using serum antibodies for confirmation.

This procedure is standard for diagnostic labs and can also be performed with commercially available kits following similar steps to ensure accurate blood grouping for clinical or educational purposes

7. VERIFICATION OF THE MENDELIAN LAWS OF INHERITANCE USING COLOURED BEADS. OBSERVATION ON GENETIC TRAITS.



1. Law of Dominance:

- Use two different colored beads to represent dominant and recessive alleles (e.g. red and white beads).
- Combine one red bead (dominant allele) and one white bead (recessive allele) to represent a heterozygous genotype.
- The resulting "offspring" will all display the dominant red color, verifying the law of dominance.

2. Law of Segregation:

- Start with a homozygous genotype (e.g. all red beads).
- "Reproduce" by randomly selecting one bead from the "parent" to represent each gamete.
- The "offspring" will show a 1:1 ratio of red and white beads, demonstrating the segregation of alleles during gamete formation.

3. Law of Independent Assortment:

- Use two sets of bead colors to represent two different traits (e.g. red/white and blue/yellow).
- Combine the beads to represent a dihybrid cross (e.g. red-blue, red-yellow, white-blue, white-yellow).
- The "offspring" will show a 9:3:3:1 phenotypic ratio, verifying the independent assortment of the two traits.

SPOTTERS

8.CELL TYPES COLUMNAR EPITHELIUM, SQUAMOUS EPITHELIUM CELL STRIATEDMUSCLE CELL & CARDIAC MUSCLE CELL

Columnar Epithelium Cell Types

- **Simple Columnar Epithelium:**
 - Tall, rectangular (column-like) cells with oval nuclei located basally.
 - **Ciliated Columnar:** Have cilia on the apical surface; help move substances over the epithelium (e.g., fallopian tubes) and **Non-ciliated Columnar:** Have microvilli (for absorption, as in the intestine); often interspersed with goblet cells that secrete mucus.
- **Simple Squamous Epithelium:**
 - Single layer of flat, wide, thin cells.
 - Located in areas specializing in passive diffusion (e.g., alveoli, capillaries).
- **Striated Muscle Cell Types**
 - Long, cylindrical, multinucleated cells (muscle fibers).
 - Contain striations due to organized sarcomeres (actin and myosin).
- **Cardiac Muscle Cell Types**
 - Striated, branched cells joined by intercalated discs.
 - Generally, contain a single central nucleus.
 - Cell types: **Myocardial Contractile Cells:** Bulk of heart muscle, responsible for contraction and blood pumping and **Myocardial Conducting Cells (Pacemaker cells):** Specialized for electrical impulse conduction (e.g., in SA node, AV node, Purkinje fibres); less contractile.

9.LIGHT MICROSCOPE

LIGHT MICROSCOPE

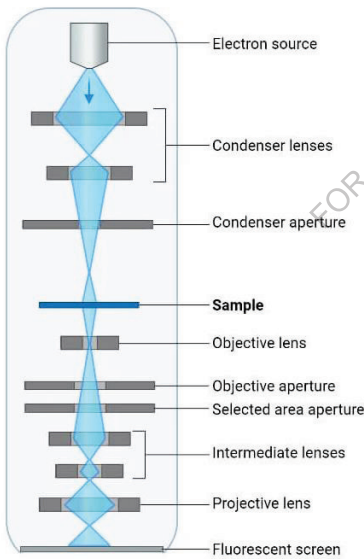
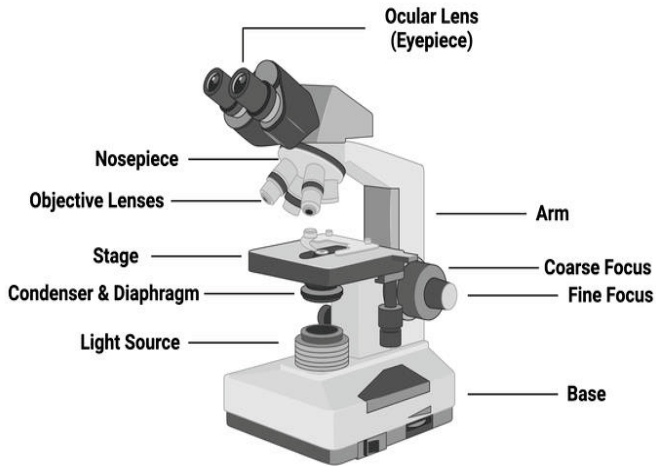
- **Light microscope** is best for observing live or fixed specimens at low resolution and moderate magnification; simple specimen prep and colour imaging are advantages.
- Two lenses which include the **objective lens** and the **eyepiece or ocular lens**.
- Objective lens is made up of six or more glasses, which make the image clear from the object
- The condenser is mounted below the stage which focuses a beam of light onto the specimen. It can be fixed or movable, to adjust the quality of light, but this entirely depends on the microscope.
- They are held together by a sturdy metallic curved back used as an **arm** and a stand at the bottom, known as the **base**, of the microscope. The arm and the base hold all the parts of the microscope.
- The stage where the specimen is placed, allowing movement of the specimen around for better viewing with the flexible knobs and it is where the light is focused on.
- Two focusing knobs i.e the fine adjustment knob and the coarse adjustment knob, found on the microscopes' arm, which can move the stage or the nosepiece to focus on the image. the sharpen the image clarity.
- It has a **light illuminator** or a **mirror** found at the base or on the microbes of the nosepiece.

TEM

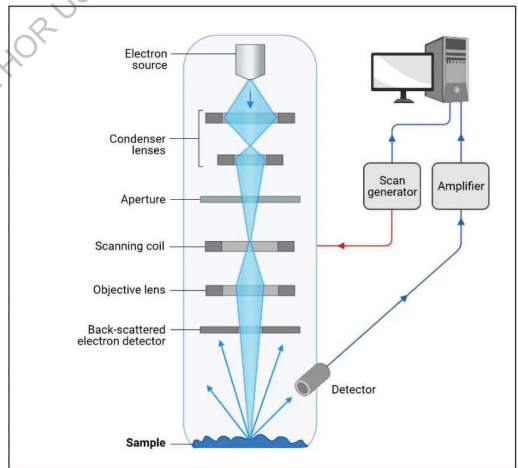
- The transmission electron microscope is used to view thin specimens through which electrons can pass generating a projection image.
- The TEM is analogous in many ways to the conventional (compound) light microscope.
- TEM is used, among other things, to image the interior of cells (in thin sections), the structure of protein molecules (contrasted by metal shadowing), the organization of molecules in viruses and cytoskeletal filaments (prepared by the negative staining technique), and the arrangement of protein molecules in cell membranes (by freeze-fracture).

SEM

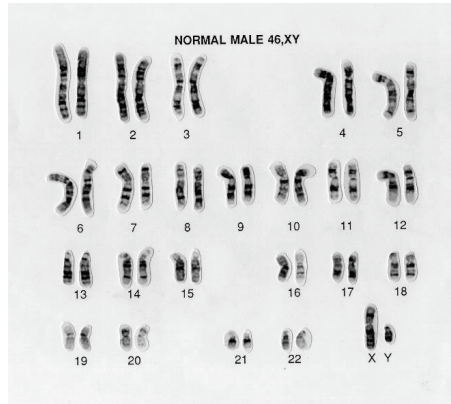
- Conventional scanning electron microscopy depends on the emission of secondary electrons from the surface of a specimen.
- Because of its great depth of focus, a scanning electron microscope is the EM analog of a stereo light microscope.
- It provides detailed images of the surfaces of cells and whole organisms that are not possible by TEM. It can also be used for particle counting and size determination, and for process control.
- It is termed a scanning electron microscope because the image is formed by scanning a focused electron beam onto the surface of the specimen in a raster pattern.



Scanning Electron Microscopy (SEM)



10. KARYOTYPING NORMAL MALE & FEMALE



NORMAL MALE KARYOTYPE

- Denoted as 46, XY.
- The 23rd chromosome pair is composed of one X and one Y chromosome.
- The X chromosome is larger, and the Y chromosome is smaller.
- This combination determines male sex characteristics.
- Useful for identifying male sex chromosome abnormalities like Klinefelter syndrome (extra X chromosome) or XYY syndrome.

NORMAL FEMALE KARYOTYPE

- Denoted as 46, XX.
- The 23rd chromosome pair consists of two X chromosomes.
- Both X chromosomes are similar in size and gene content.
- This configuration determines female sex characteristics.
- Used to identify female chromosome abnormalities such as Turner syndrome (missing one X) or triple X syndrome (an extra X chromosome).

**11. KARYOTYPING FINDINGS IN DIFFERENT CHROMOSOMAL SYNDROMES,
HIGHLIGHTING THEIR CHROMOSOMAL ABNORMALITIES AND CHARACTERISTIC
KARYOTYPES:**

1. Down Syndrome

- Trisomy 21 (three copies of chromosome 21)
- 47,XX,+21 or 47,XY,+21
- Most common trisomy; causes intellectual disability, characteristic facial features, heart defects.

2. Patau Syndrome

- Trisomy 13
- 47,XX,+13 or 47,XY,+13
- Severe intellectual disability, cleft lip/palate, heart defects, usually fatal early.

3. Turner Syndrome

- Monosomy X (missing second sex chromosome)
- 45,X or 45,X0
- Female phenotype, short stature, infertility, heart defects.

4. Klinefelter Syndrome

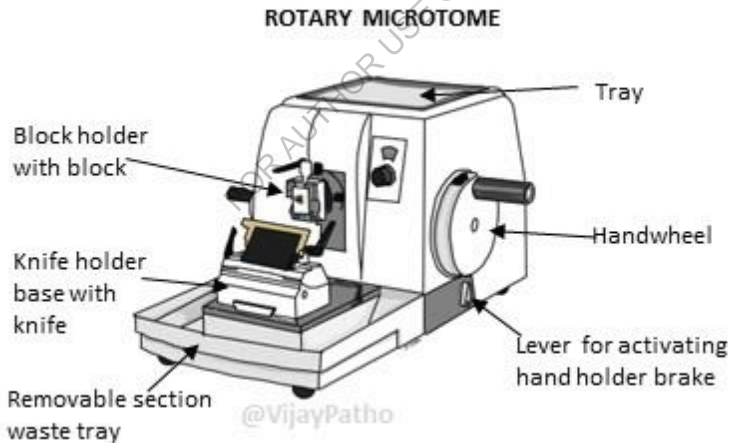
- Extra X chromosome in males
- 47,XXY
- Male phenotype; tall stature, learning difficulties, infertility.



12. TYPES OF MICROTOMES, SECTIONING OF PARAFFIN BLOCKS. STAINING OF PARAFFIN SECTIONS.

TYPES OF MICROTOMES

- I. **Rotary Microtome:** This is the most commonly used microtome in routine histopathology. It operates by rotating a sample holder toward a fixed knife, allowing for precise sectioning of paraffin-embedded tissues at thicknesses ranging from 1 to 60 μm . The tissue must be adequately hardened through dehydration and embedding in paraffin before sectioning.
- II. **Sledge Microtome:** This microtome features a fixed holder that moves across a knife, allowing for the cutting of larger samples. It is less common but still used for specific applications.



13. EMBEDDING –SECTIONING OF PARAFFIN BLOCKS

1. **Tissue Preparation:** Tissues are fixed, dehydrated, and cleared before being embedded in melted paraffin. Once cooled, the paraffin forms a solid block around the tissue.
2. **Cutting:** The paraffin block is then mounted on a microtome, where it is sliced into thin sections. The thickness can vary based on the type of microtome used and the requirements of the analysis, typically between 2 to 50 μm .
3. **Mounting:** The sections are mounted on microscope slides for further processing.



14. FIXATIVE AND BOVINE SERUM IN HISTOLOGY

- BSA is also widely used as a blocking agent in immunohistochemistry and immunofluorescence to prevent nonspecific binding of antibodies, thus reducing background noise.
- While not a fixative, BSA's role in preparing specimens ensures improved signal specificity during staining.
- In some protocols, BSA is combined with detergents (like Triton X-100) during antibody incubation steps.

15.



STAINING OF HEAMATOXYLINE AND EOSIN STAIN

- Haematoxylin is a basic dye that stains *basophilic* structures (primarily nuclei containing DNA and ribosomal RNA) blue to purple.
- Eosin is an acidic dye that stains *acidophilic* structures (cytoplasm, extracellular matrix, muscle fibres, red blood cells) pink to red/orange. The contrast allows clear differentiation between nuclear and cytoplasmic elements under a light microscope.



16. THE EBONY AND YELLOW GENES IN *DROSOPHILA MELANOGASTER* PLAY CRUCIAL ROLES IN DETERMINING BODY COLOR AND MELANIN PATTERNS.

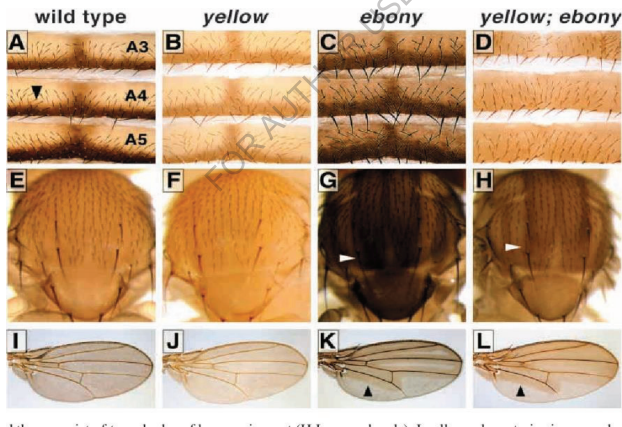
BODY COLOR MUTANT

a) EBONY MUTANTS

- ebony mutants, such as ebony1 (e1), block the production of yellow NBAD sclerotin pigment.
- In ebony mutants, more dopamine is available to produce black and brown melanin, resulting in a strongly melanized, dark cuticle.
- ebony mutants have a temperature increase of around $0.6^{\circ}\text{C} \pm 0.2^{\circ}\text{C}$ compared to wild-type flies when exposed to light, due to their darker pigmentation.

b) YELLOW MUTANTS

- yellow mutants, such as yellow1 (y1), cannot produce black melanin.
- yellow mutants have a cuticle pigmented only with brown melanin and yellow NBAD sclerotin, resulting in a lighter body colour.
- yellow mutants have a temperature increase of around $0.6^{\circ}\text{C} \pm 0.2^{\circ}\text{C}$ less compared to ebony mutants when exposed to light, due to their lighter pigmentation.



WINGS MUTANT

a) CURLY WING MUTANT

- The **Curly** (Cy) mutation is a dominant trait located on the second chromosome.
- It causes the wings to bend upwards, creating a characteristic curled appearance.
- The mutation is associated with two large inversions in the chromosome and is known to be lethal in homozygous form, meaning that individuals with two copies of the curly allele do not survive.

b) VESTIGIAL WING MUTANT

Genetic Basis

- The **Vestigial** (vg) mutation is another wing mutant that results in significantly reduced wing size or complete wing absence. It is also located on the second chromosome and is recessive.



EYE COLOR MUTANT

c) Bar Eye Mutant

- The Bar (B) mutation is a dominant mutation located on the X chromosome that causes the eyes to be reduced in size and kidney-shaped.
- Bar is caused by a chromosomal rearrangement and exhibits a position effect, where the eye colour is variable.

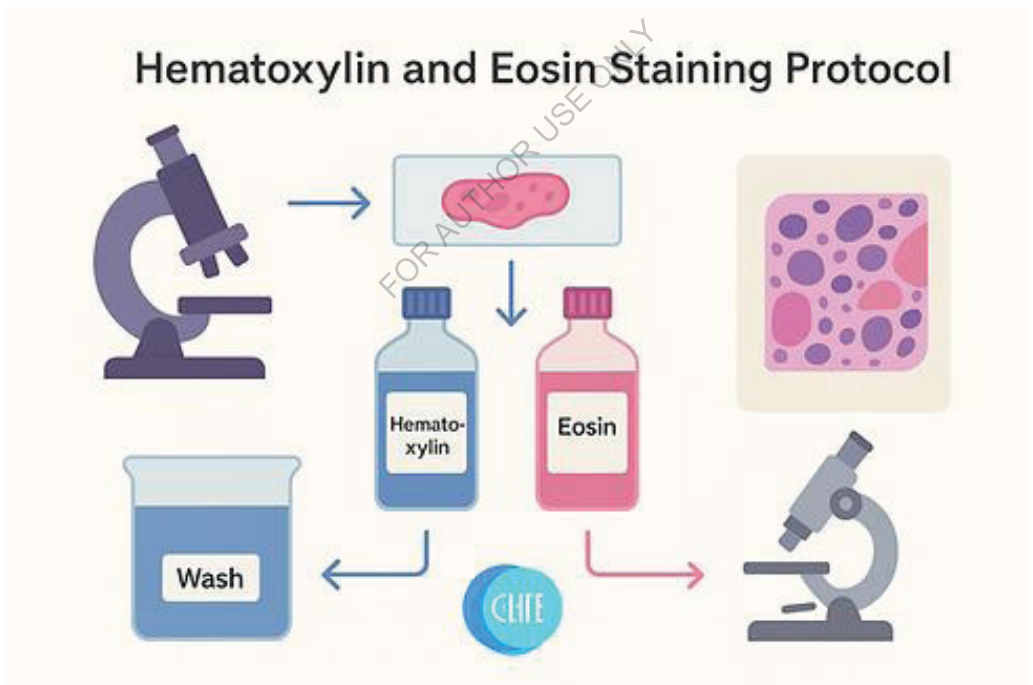
d) White Eye Mutant

- The white (w) gene is located on the X chromosome and is required for the production of red and brown eye pigments (pteridines and monochromes).
- white mutants have white eyes due to the lack of both pteridine and monochrome pigments.
- white is commonly used as a visible marker for genetic transformation in insects.

17. METHODS OF HAEMATOXYLIN AND EOSIN STAINING

Principle of H&E Staining

- **Haematoxylin:** This basic dye primarily stains nucleic acids, imparting a blue-purple colour to the nuclei of cells. It binds to acidic structures within the cell, such as DNA and RNA, due to its positive charge.
- **Eosin:** An acidic dye, eosin stains the cytoplasm and extracellular matrix, resulting in varying shades of pink, orange, and red. It interacts with basic proteins, highlighting structures like red blood cells and collagen.



18. SLIDES OF MAMMALIAN SPERM AND OVUM.

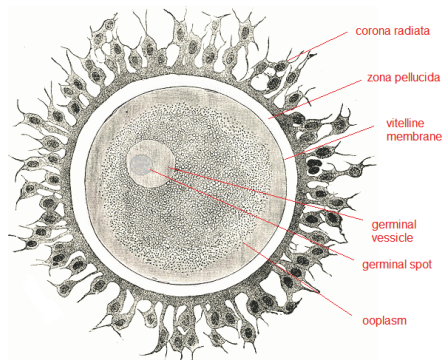
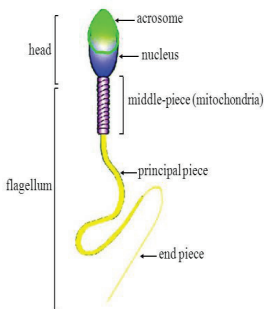
Mammalian Sperm Slides

- Mammalian sperm are minute, microscopic, flagellated, and motile gametes with no nutritive material or protective envelopes.
- Prepared microscope slides of mammalian sperm smears are available that show the characteristic features of sperm cells.
- These slides are useful for biology classrooms to explore the structure-function relationship of sperm as per NGSS standards.
- The slides are professionally stained and prepared, with the sperm cells oriented and characterized based on the direction of tail bending relative to the head midline.
- Measurements such as waveform amplitude, microtubule sliding, and curvature can be extracted from the sperm flagellum to study the "sliding filament" hypothesis of sperm motility.

Mammalian Ovum Structure

- The mammalian ovum is a spherical, non-motile gamete with a large amount of cytoplasm.
- The cytoplasm is differentiated into an outer, smaller, and transparent exoplasm (or egg cortex) and an inner, larger, and opaque endoplasm (or ooplasm).
- The nucleus of the ovum is large and eccentric in position, called the germinal vesicle. This gives the ovum polarity, with the animal pole containing the nucleus and polar body, and the opposite vegetal pole.
- The ovum is surrounded by several membrane layers:
 - Primary membrane: Zona pellucida, secreted by the egg itself
 - Secondary membrane: Corona radiata, contributed by the ovarian follicle cells
 - Tertiary membrane: Secreted by the reproductive tract, not present in mammals
- The ovum contains yolk (nutritive material) synthesized through the process of vitellogenesis, composed mainly of proteins and phospholipids.

MAMMALIAN SPERM



19.SLIDES OF DIFFERENT CLEAVAGE STAGES – BLASTULA, GASTRULA AND MORULA OF FROG.

MORULA STAGE

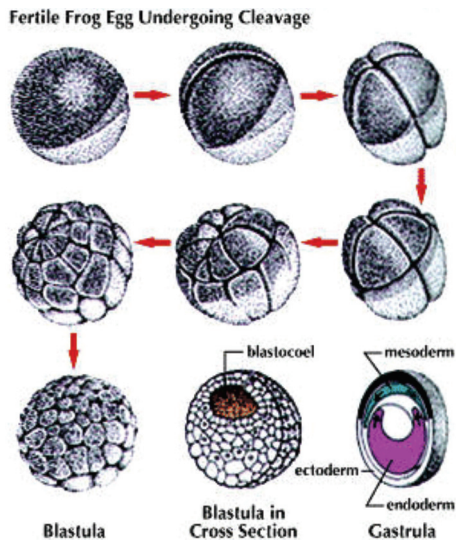
- The cells formed by cleavage are called blastomeres.
- The upper pigmented blastomeres are called micromeres, while the lower yolky ones are macromeres.
- Further cleavages divide the micromeres more rapidly than the macromeres.
- The blastomeres' mutual pressure flattens their surfaces in contact with each other.
- At this stage, the embryo acquires a mulberry-like appearance, hence the name "morula".

BLASTULA STAGE

- The blastula has a fluid-filled cavity called the blastocoel, located above the equator in an eccentric position.
- The blastocoel is surrounded by two types of blastomeres - micromeres at the animal pole and macromeres at the vegetal pole.
- The blastula is a hollow ball of cells with a fluid-filled cavity.

GASTRULA STAGE

- During gastrulation, the three primary germ layers (ectoderm, mesoderm, endoderm) are formed.
- The ectoderm is made up of micromeres and surrounds the embryo.
- The endoderm is well developed and internalized.
- The mesoderm is formed from the roof of the archenteron (primitive gut) and the floor of the archenteron.
- The yolk-laden macromeres are pushed towards the blastopore, forming the yolk plug.



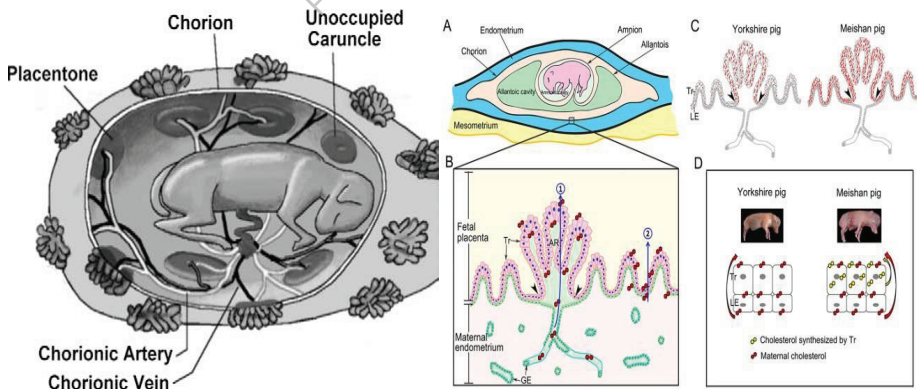
20. PLACENTA OF SHEEP AND PIG.

SHEEP PLACENTA

- The sheep placenta is cotyledonary, meaning it has discrete button-like structures called cotyledons that are the foetal portion of the placentomes.
- There are 70-100 cotyledons in the sheep placenta, with the number and size varying based on maternal age and breed.
- The cotyledons are covered by trophoblast and foetal villi that interdigitate with the maternal caruncles to form the placentomes.
- The intercotyledonary chorion has absorptive areolae between the cotyledons.
- The sheep placenta is epitheliochorial, with six tissue layers separating maternal and foetal blood.
- Syncytial plaques containing 20-24 nuclei are extensive in the sheep placenta and persist throughout pregnancy.

PIG PLACENTA

- The pig placenta is diffuse, meaning the chorionic villi are evenly distributed over the entire chorion.
- Pigs have an epitheliochorial placenta like sheep, with six tissue layers between maternal and foetal blood.
- The pig placenta is less invasive compared to some other species, with the maternal tissues remaining intact.
- The chorionic villi in the pig placenta are simple in structure, with a single layer of trophoblast covering the villi



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Focuses on Mendelian genetics, chi-square, linkage mapping, cytology techniques; suitable for B.Sc./B.Sc. (Agri) level.
23. **“Principles of Genetics and Cytogenetics – BSH 115 Practical Manual” – Rajeshkumar Singh & Diana Sagolsem**
Indian agricultural university manual with 13 core practicals: fixatives and stains, mitosis/meiosis, chi-square, three-point test cross, linkage map construction.
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NEP-aligned agriculture/genetics practical manual; covers basic Mendelian experiments and allied exercises for early semesters.
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Combined cell biology–genetics manual (karyotyping, Mendelian crosses, cytology). Useful as a template for designing integrated practicals.
26. **Broader genetics & cell biology handbooks**
27. **“A Practical Hand Book of Genetics and Cell Biology” (2025)**
Designed for B.Sc. students in Indian universities, aligned with NEP-2020; includes mitosis, meiosis, population genetics, karyotyping; has solved/unsolved problems.
28. **“Genetics Laboratory Manual” – Babcock & Collins (McGraw-Hill, classic text)**
Historical but still useful for classical genetic analysis exercises.
29. **Molecular genetics–oriented manuals**
30. **“Genetics Laboratory Manual – BIO2450L” (City Tech, CUNY)**
Modern undergraduate manual with labs on probability, karyotyping, recombination, PCR, genotyping, GMO analysis, and DNA barcoding.

31. “Genetics Laboratory Manual” (molecular lab skills manual)

Covers micropipetting, centrifugation, agarose gels, PCR, restriction digests, transformation (pGLO), and Alu insertion polymorphism.

32. “Laboratory Manual for Molecular Biology” – Veerashaiva College, Ballari

B.Sc. NEP molecular biology manual: DNA quantification, electrophoresis, etc., useful to complement classical genetics practical.

33. Genetics + plant breeding / applied focus

34. “Practical Manual of Genetics & Plant Breeding” – Bineeta Singh & G.M. Lal

Indian agriculture-oriented practical manual; covers genetic principles plus plant breeding exercises.

35. “Practical Guide to Genetics, Plant Breeding & Seed Technology” – Akshay Talukdar & Anil Kumar Chaudhary

Book-length guide with genetics fundamentals plus applied breeding and seed technology.

36. A Practical Guide to Developmental Biology” – Melissa A. Gibbs (Oxford University Press)

Book-length lab manual introducing many model systems with classical observations, immunohistochemistry, and gene-expression manipulation; aimed at upper-undergraduate/graduate labs.

37. “Key Experiments in Practical Developmental Biology” (collection described within the same guide)

Presents 27 structured exercises with discussion of difficulty, common pitfalls, and timing, designed specifically for teaching practical developmental biology.

38. “Laboratory Exercises in Developmental Biology” – Cruz

Series of structured classical embryology labs widely used as teaching exercises.

39. Course-specific university manuals

40. BZ311 Developmental Biology Lab Manual (Colorado State University)

Modern lab manual integrating experimental and observational labs, often including sea urchin fertilization and vertebrate embryo work.

41. BIO 351L Developmental Biology Labs (Reed College)

Set of PDF lab protocols: sea urchin and zebrafish development, gene expression studies, and experimental perturbations.

42. *ZOO Developmental Biology Lab Manual (Univ. of Guelph) **

Includes labs on histology, gametogenesis, sea urchin fertilization vs. environmental factors, frog/urchin development, with quantitative data analysis and report writing

43. Human embryology–focused practical resources

44. Embryology Lab Guide – University of Maryland School of Medicine

Lab guide organized around preserved abnormal human development cases, emphasizing developmental defects and their embryological basis.

45. Inderbir Singh’s “Human Embryology” with lab-oriented sections

Textbook for MBBS that integrates basic developmental processes, genetics, and clinical correlations; useful to design slide-based human embryology practicals.

46. Concise practical manuals/atlasses

47. “Practical Manual of Developmental Biology” (Scribd manual)

Designed to provide both text and diagrams of developing stages, addressing the gap where many manuals only give drawings without explanatory notes.

48. “A Class Book of Practical Embryology” – Odgers

Classic resource using serial sections of pig embryos to teach practical section reading and three-dimensional reconstruction in embryology.

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